

Design of a RF Doherty Power Amplifier Based on Handset Application

Yujie Tang¹, Yingjie Zhou¹, Hao Zhou¹, Lei Yang², Yuan Zheng², Yufeng Guo¹, Yi Zhang^{1,3,4*}

¹College of Integrated Circuit Science and Engineering,

Nanjing University of Posts and Telecommunications, Nanjing, 210023, China

²Nanjing GuoBo Electronics Co. Ltd, Nanjing, 210016, China

³Science and Technology on Monolithic Integrated Circuits and Modules Laboratory, Nanjing, 210016, China

⁴State Key Laboratory of Millimeter Waves, Nanjing, 210096, China

*corresponding author: nyzhangyi@njupt.edu.cn

Abstract- A Doherty Power Amplifier (PA) with high efficiency and high linearity that can be used in smart communication devices such as mobile phones in the sub-6GHz frequency band is proposed. This Doherty PA uses the InGaP/GaAs hetero-junction bipolar transistor (HBT) process. Compared with the traditional Doherty PA, the PA in this paper combines the choke inductance of the output collector and the phase compensation circuit with the output circuit of the Doherty PA respectively, which reduces the number of components required for matching. From the simulation, when the power supply voltage of the Doherty power amplifier is 4.2V, the operating frequency is 3.5GHz and the output power is 26.5dBm, the power added efficiency (PAE) reaches 34.5%, at the same time when using a 10MHz bandwidth LTE-3GPPFDD modulated signal the Adjacent Channel Power Ratio (ACPR) below to -35dBc.

Key Words- Doherty, power amplifier (PA), hetero-junction bipolar transistors (HBT), PAE.

I. INTRODUCTION

In order to meet the development of modern communication technology and the increase of data transmission, the transmission signal is required to use a modulated signal with a large bandwidth and a peak-to-average ratio as a key component of the transmitter, so the power amplifier needs to improve the efficiency of power back-off in a wide frequency band. Doherty PAs are widely used in modern communications due to its good linearity, simple structure, and high efficiency at power back-off.

Doherty technology originated in the 1930s and is a new type of load modulation technology proposed at that time. This structure can effectively improve the efficiency of amplifiers. Currently, Doherty structures are widely used in power amplifiers^[1]. In recent years, linear Doherty PA has become very popular in mobile phones and micro base stations due to direct input power allocation and more precise matching techniques overcoming size limitations. At the same time, the power amplifier of this structure can have higher linearity through appropriate harmonic control^[2].

The authors in [3] proposed a technique for correctly allocating inputs without coupling, and introduced a matching method that can reduce the number of components used. At the

same time, the author provides a detailed explanation of the output matching method for improving linearity and efficiency under non-uniform input driving.[4] points out that the quarter-wave transformer limits the bandwidth of Doherty PA. In order to improve the bandwidth of Doherty PA, this article uses a Wilkinson power divider that has a lumped structure. The author has also verified the proposed bandwidth enhancement technology, enabling it to be applied to Doherty PA. In [5] the author designed a linear Doherty PA which has an enhanced back-off efficiency mode. In order to achieve high linearity, the author studied gain modulation and ways to reduce IMD3. The author also proposes a compact design approach to reduce the area of the chip. In the article [6], the design of a new compact Doherty power amplifier suitable for handset applications is proposed and the effectiveness of the proposed Doherty PA for handset applications is verified by the measurement results.

Fig. 1 shows the difference between the structure of traditional Doherty PA and Doherty PA in this article.

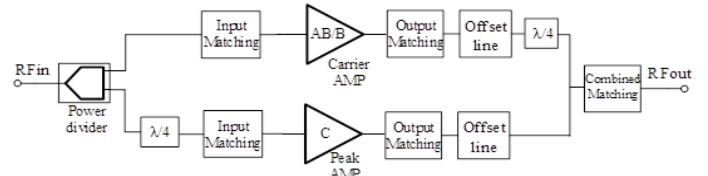


Fig. 1. (a) Traditional Doherty PA.

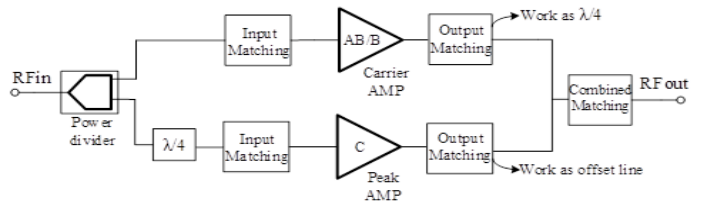


Fig. 1. (b) Proposed Doherty PA.

II. STRUCTURE OF DOHERTY PA

A. Class-C-biased peaking amplifier

The Doherty structure consists of two amplifiers: a carrier power amplifier and a peak power amplifier. These two amplifiers do not work simultaneously. The carrier power amplifier will work first, at which point the input power is small. However, the

peak power amplifier will only start working when the input power continues to increase and finally reaches the opening value of the peak power amplifier. Considering the different working modes of the two amplifiers, the carrier power amplifier usually selects Class AB or Class B biased amplifiers. However, since the gain of Class C biased amplifiers can vary with the input, the peak power amplifier considers using Class C biased amplifiers.

B. Quarter-wave transformer

On the one hand, the quarter-wave transformer located after the carrier amplifier plays an impedance matching role, which matches the output impedance of the carrier amplifier to the required value to ensure that the carrier amplifier can still have a large output current when the peak amplifier is working. On the other hand, it can also play a role in compensating for phase changes caused by different biases or some devices between two amplifiers. Ultimately, it can make the output power of two power amplifiers tend to be the same value. Considering the area, a lumped structure transformer was used to replace the traditional microstrip structure, which can reduce the circuit area.

C. Phase compensation network

Due to the phase change caused by quarter-wave transformer and other components, phase compensation needs to be considered before combining the carrier amplifier and peak amplifier, and the input matching circuit precisely has the function of phase compensation.

In Fig. 2, in order to compensate for the phase issues caused by the quarter-wave transformer and different biases, two high pass filters are used in the carrier amplifier circuit, which also serves as a matching circuit, while the peak amplifier uses a bandpass filter.

D. Output collector choke inductance

Because additional offset lines in traditional Doherty reduce bandwidth, this paper incorporates them into the output matching process. In this article, The bias line not only compensates for the inherent capacitance of the transistor in the circuit, but also plays a role in output matching to modulate the output load to the desired size. In the proposed structure, carrier amplifier's output capacitance is combined with the capacitance in the lumped quarter-wave transformer, while for the peak amplifier, the output capacitance is resonant through the choke inductance of the collector.

III. SIMULATION AND LAYOUT

Fig. 3 is a small-signal S-parameter simulation diagram when the power supply voltage is 4.2V. In the figure, the input scattering coefficient S11 and the output scattering coefficient S22 are both less than -10dB, which indicates that the Doherty PA has good matching performance and the small signal gain S21 of the amplifier is flat and stable from 3.4GHz to 3.6GHz, and its value is about 30.5dB.

As shown in Fig. 4.(a) and Fig. 4.(b) are the curves of output gain, output power and input power, respectively. During the operation of the amplifier, the gain tends to be stable. Before the peak power amplifier is turned on, the gain is mainly contributed by the carrier power amplifier, which is stable at about 30dB; When the input power is -10dBm, the peak power amplifier starts to work, at this time, the carrier power amplifier reaches saturation and the gain reaches the maximum, and then the peaking power amplifier compensates for the drop in carrier power amplifier gain, so that the overall gain remains stable.

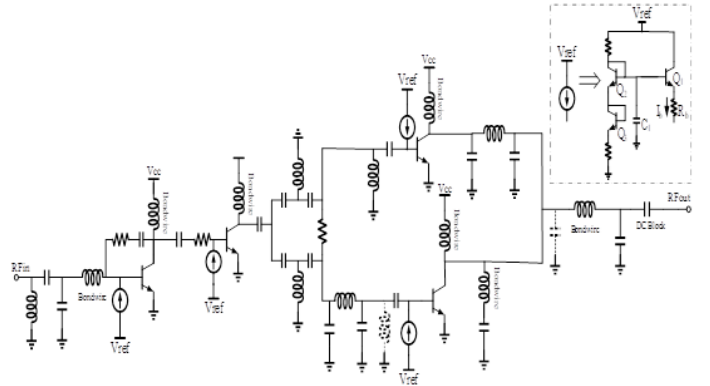


Fig. 2. Schematic of Doherty PA.

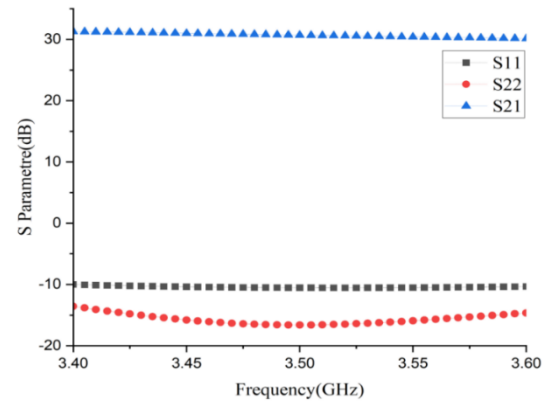


Fig. 3. Simulation of Small-signal S-parameter.

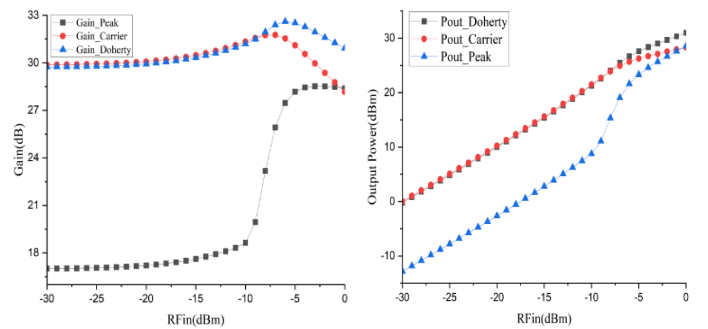


Fig. 4. (a) Curves of Output gain.

(b) Curves of Output power.

Fig. 5 is simulation of PAE, Gain, and IMD3. IMD3 adopts HB two-tone signal simulation method, centered on 3.5GHz, and the frequency span is 1MHz. When the output power is 26.5dBm, the PAE of the PA reaches 34.5%. After using the Doherty structure, the PA has a great IMD3 curve.

Fig. 6 shows the simulation of the ACPR, when the center frequency is 3.5GHz and the output power is 26.5dBm, the ACPR of the LTE-3GPPFDD modulated signal with a bandwidth of 10MHz is below -35dBc.

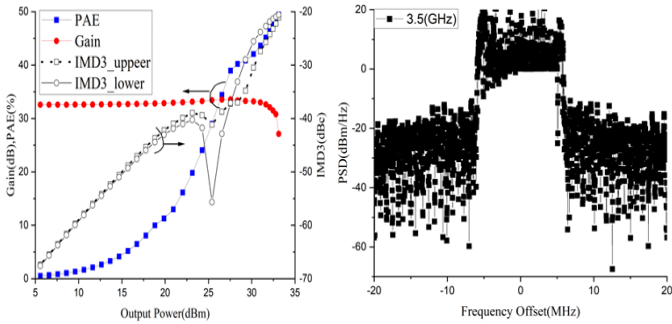


Fig. 5. Simulation of PAE, Gain, IMD3. Fig. 6. Simulation of ACPR.

The Doherty PA in this article uses the InGaP/GaAs HBT process. Fig. 7 shows the layout design of the Doherty PA with the size of $1.4 \times 1.7 \text{ mm}^2$.

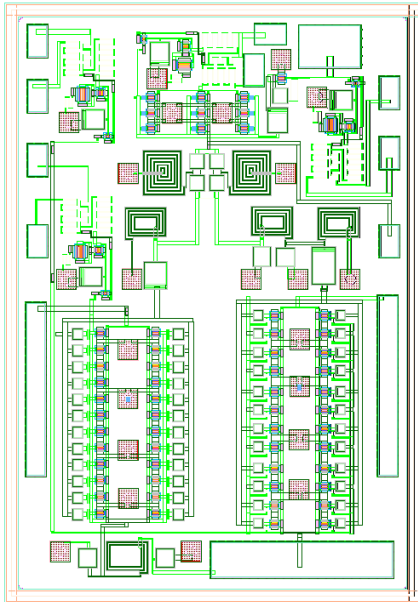


Fig. 7. Layout design.

IV. CONCLUSION

In this paper, a $1.4 \times 1.7 \text{ mm}^2$ Doherty Power Amplifier is designed by using InGaP/GaAs HBT process. From the simulation results, when the power supply voltage of the Doherty PA is 4.2V, the center frequency is 3.5GHz and the output power is 26.5dBm, the PAE reaches 34.5%. The ACPR of a 10MHz bandwidth LTE-3GPPFDD modulated signal below -35dBc.

ACKNOWLEDGMENT

This work was supported by National Natural Science Foundation of China (61804081), The Open Foundation of Science and Technology on Monolithic Integrated Circuits and Modules Laboratory (614280304012101), China Postdoctoral Science Foundation (2018M642292), Open project of State Key Laboratory of Millimeter Waves (K202220), Natural Science Foundation of Nanjing University of Posts and Telecommunications (NY221065) and in part by postgraduate research & practice innovation program of jiangsu province(SJCX22_0262).

REFERENCES

- [1] D. Kang, Y. Cho, D. Kim, B. Park, J. Kim and B. Kim, "Impact of Nonlinear C_{bc} on HBT Doherty Power Amplifiers," in *IEEE Transactions on Microwave Theory and Techniques*, vol. 61, no. 9, pp. 3298-3307, Sept. 2013.
- [2] Y. Cho, K. Moon, J. Kim, B. Park and B. Kim, "Linear Doherty power amplifier with adaptive bias circuit for average power-tracking," *2016 IEEE MTT-S International Microwave Symposium (IMS)*, 2016, pp. 1-3.
- [3] D. Kang, J. Choi, D. Kim and B. Kim, "Design of Doherty Power Amplifiers for Handset Applications," in *IEEE Transactions on Microwave Theory and Techniques*, vol. 58, no. 8, pp. 2134-2142, Aug. 2010.
- [4] D. Kang *et al.*, "1.6–2.1 GHz broadband Doherty power amplifiers for LTE handset applications," *2011 IEEE MTT-S International Microwave Symposium*, 2011, pp. 1-4.
- [5] Y. Cho, D. Kang, J. Kim, K. Moon, B. Park and B. Kim, "Linear Doherty Power Amplifier With an Enhanced Back-Off Efficiency Mode for Handset Applications," in *IEEE Transactions on Microwave Theory and Techniques*, vol. 62, no. 3, pp. 567-578, March 2014.
- [6] K. Takenaka, T. Sato, H. Matsumoto, M. Kawashima and N. Nakajima, "New compact Doherty power amplifier design for handset applications," *2017 IEEE Topical Conference on RF/Microwave Power Amplifiers for Radio and Wireless Applications (PAWR)*, 2017, pp. 81-83.